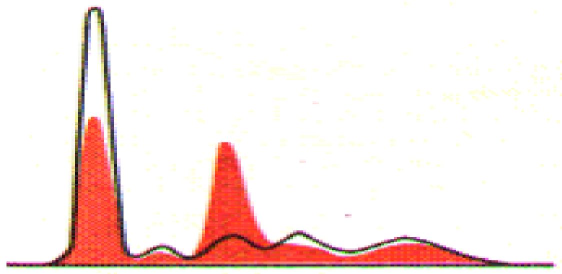
CPR IMCQ 10

KEY

Fall 2025

1. A 7-year-old girl is brought to the physician by her mother because of a 1-week history of progressive swelling of her feet. The mother says that her daughter felt tired, had low appetite but had gained weight. Physical examination shows puffy eyelids and bilateral pitting pedal edema. Serum protein electrophoresis is shown. Which of the following best explains the edema in this patient?

A. Reduced albumin synthesis by liver	14%
B. Increased alpha-2 macroglobulin	14%
C. Increased neutrophil elastase activity	14%
D. Decreased biliary copper excretion	14%
E. Reduced capillary oncotic pressure	14%
F. Reduced dietary protein content	14%
G. Increased serum LDL-cholesterol	14%



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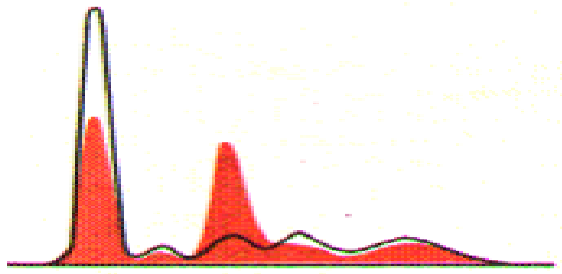

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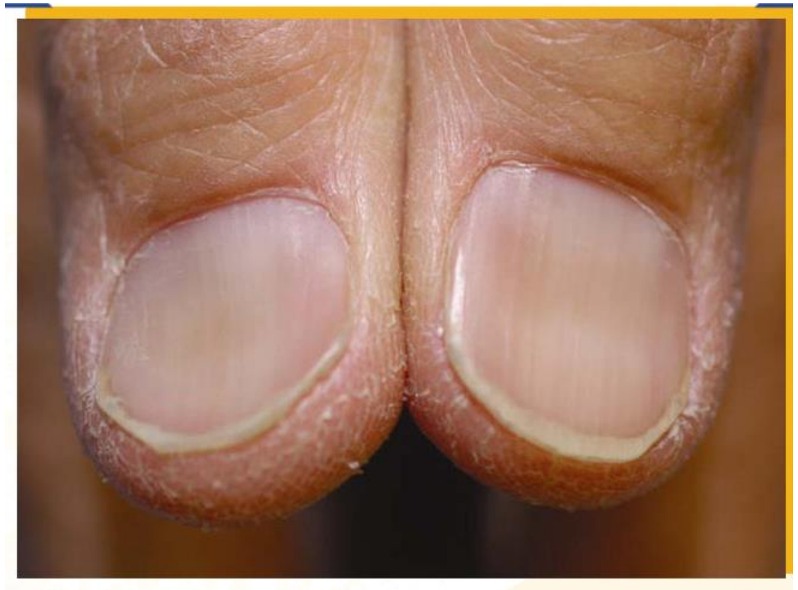
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0319; Predict three common causes of hypoalbuminemia (liver disease, nephrotic syndrome, protein malnutrition) and explain the biochemical basis for the occurrence of edema in hypoalbuminemia.

2. A 42-year-old woman comes to the physician because of 2-week history of dizziness. She has a history of heavy, irregular periods occurring every 2-3 weeks of 5-7 days duration and became heavier with her last period lasting "half a month". Her pulse is 87/min, respirations are 17/min, blood pressure is 107/47 mmHg and temperature is 36.1°C (96.9°F). Physical examination shows pale mucous membranes and conjunctiva, spooning of her nails and a grade 2 systolic murmur. Laboratory studies are shown in the table. A peripheral smear shows microcytic hypochromic anemia. Which of the following additional laboratory findings are most likely in this patient?

	Patient	Reference
Hemoglobin	6 g/dL	12-16 g/dL
Hematocrit	15.8%	36-44%
Serum iron	40 µg/dL	50-170 µg/dL
Folate	12.8 ng/mL	≥ 4.6 ng/mL
Vitamin B12	1,019 pg/mL	232-1245 pg/mL



A. Low haptoglobin levels 17%
B. Increased hemopexin levels 17%
C. Low transferrin saturation 17%
D. Increased alpha-2 macroglobulin levels 17%
E. Low albumin levels 17%
F. Low ceruloplasmin levels 17%

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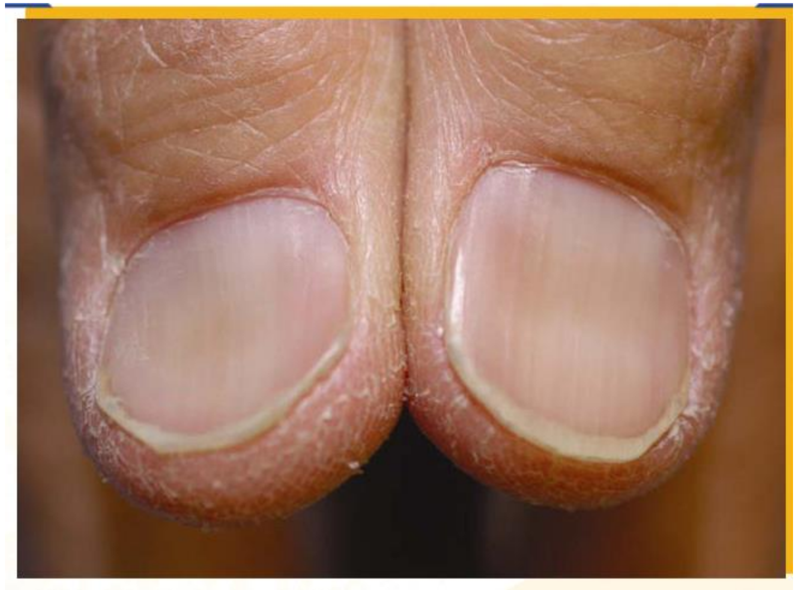

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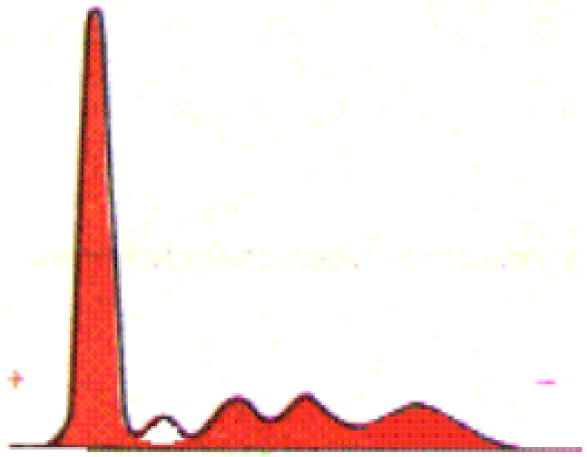
✓ C. Low transferrin saturation
D. Increased alpha-2 macroglobulin levels
E. Low albumin levels
F. Low ceruloplasmin levels

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0322; Distinguish the functions of proteins found in the beta-globulin fraction (transferrin, hemopexin, beta lipoprotein).

3. A 40-year-old man comes to the physician because of a 6-month history of progressive shortness of breath. He has had a cough productive of white sputum for 2 years. He smoked 1 pack of cigarettes daily for 16 years but quit 10 years ago. He is in mild respiratory distress with pursed lips and a barrel chest; he is using the accessory muscles of respiration. Breath sounds are distant, and crackles are present in the lower lung fields bilaterally. Pulmonary function tests show a decreased FEV1:FVC ratio, increased residual volume, and decreased diffusion capacity. An x-ray of the chest shows hyperinflation and hyper translucency of the lower lobes of both lungs. Serum protein electrophoresis is shown. Which of the following best describes the deficient serum protein?

A. Synthesized in the lung and secreted into the blood — 20%
B. Lung pathology is due to attainment of a "novel function" — 20%
C. Binds to and inhibits the action of a serine protease — 20%
D. Encoded by the neutrophil elastase gene — 20%
E. The pathogenic allele is the "M" allele — 20%



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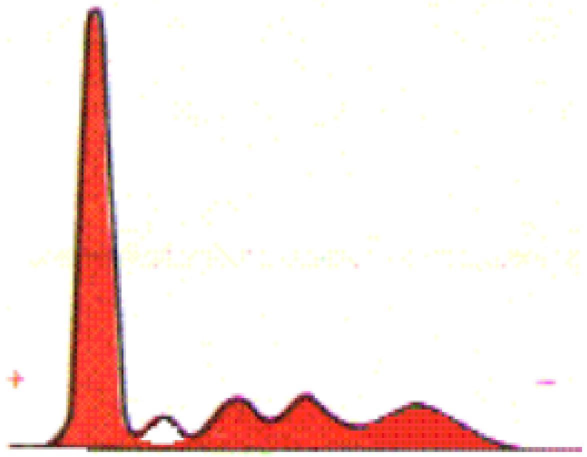
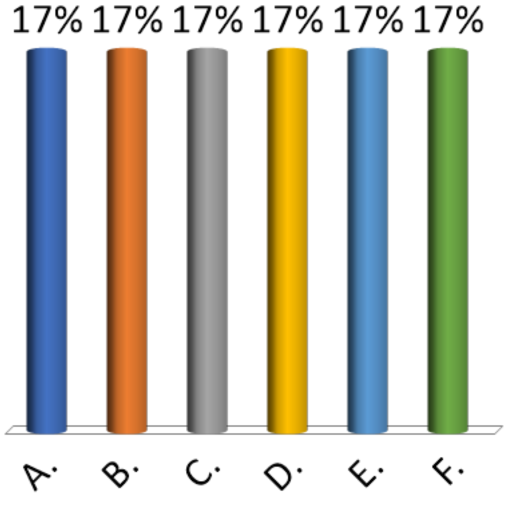


SOM.MK.I.BPM1.1.CPR.2.BCHM. 0312; Discuss the function of the protease inhibitor alpha 1 antitrypsin (AAT) encoded by the SERPINA1 gene.

4. A 52-year-old man comes to the physician for a follow-up examination. He has a 10-year history of hypertension and is taking anti-hypertensive medications. His blood pressure is 132/84 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. Results of urine studies are within reference ranges. Results of renal function studies show a renal plasma flow (RPF) of 550 mL/min and a filtration fraction (FF) of 0.20. Which of the following values best approximates the glomerular filtration rate (GFR) in this patient?

A. 75 mL/min
B. 90 mL/min
C. 100 mL/min
D. 110 mL/min
E. 120 mL/min
F. 140 mL/min



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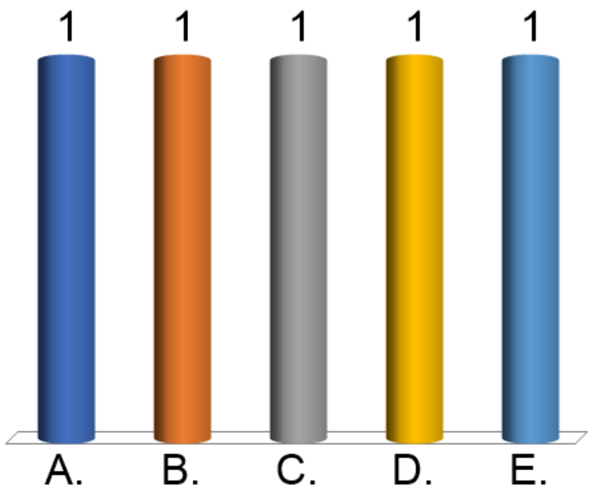
FF = GFR/RPF
GFR = RPF x FF
= 550x0.20 = 110

SOM.MK.II.BPM1.3.CPR.2.PHYS.0284. Define renal blood flow, renal plasma flow, glomerular filtration rate, and filtration fraction and list typical values.

5. A 71-year-old man comes to the physician because of a 6-week history of reduced urine volume, swollen feet, and shortness of breath. He has a 20-year history of type 2 diabetes mellitus. His blood pressure is 144/90 mm Hg; other vital signs are within normal limits. Physical examination shows pitting edema of feet. Auscultation of the lungs shows coarse crackles in the bases of both lungs. Laboratory studies show elevated serum creatinine, blood urea, and reduced glomerular filtration rate. Examination of a renal biopsy specimen shows thickening of the glomerular basement membrane and reduced permeability. He is diagnosed with diabetic nephropathy. Which of the following mechanisms best explains a decrease in glomerular filtration rate in this patient?

A. Decrease in filtration coefficient
B. Increase in hydrostatic pressure in Bowman space
C. Decrease in Bowman space oncotic pressure
D. Decrease in plasma oncotic pressure
E. Decrease in hydrostatic pressure in glomerular capillary



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C. Decrease in Bowman space oncotic pressure
D. Decrease in plasma oncotic pressure
E. Decrease in hydrostatic pressure in glomerular capillary

SOM.MK.II.BPM1.3.CPR.2.PHYS.0277. Define the filtration coefficient at the glomerular capillary, describe the membrane properties that contribute to it, and explain its role in determining GFR.

6. A 26-year-old healthy man participates in a study that evaluates the effect of a new drug on the renal blood flow and GFR. This drug selectively dilates efferent arterioles in the kidneys. The renal blood flow and glomerular filtration rate are measured before and after administration of the drug. Which of the following sets of alterations is most likely in this participant after administration of the drug?

Option	Glomerular filtration rate (GFR)	Renal plasma and blood flow	Filtration fraction (FF)	
A	Decreased	Increased	No change	0
B	Increased	Decreased	Decreased	0
C	Increased	Decreased	Increased	0
D	Decreased	Increased	Decreased	0
E	Increased	Increased	No change	0

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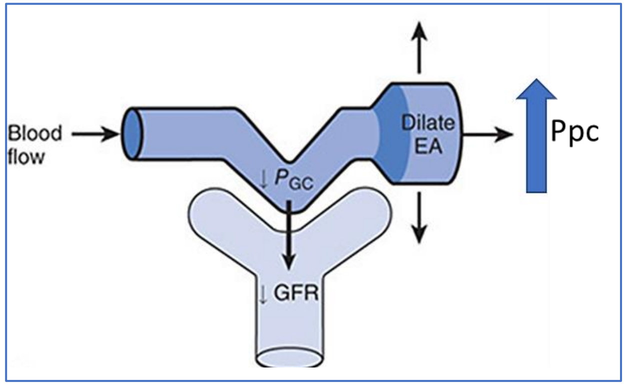
E	Increased	Increased	No change	0

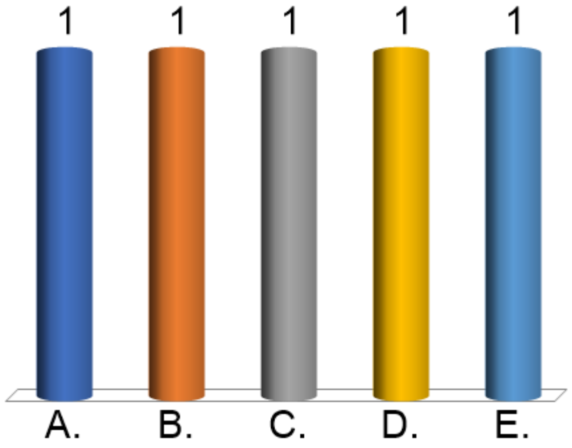


FF = GFR/RPF
= ↓GFR/↑RPF
= ↓ in FF

SOM.MK.II.BPM1.3.CPR.2.PHYS.0279. Describe the relative resistances of the afferent and efferent arterioles and the effects on renal blood flow, GFR and FF of selective changes in each.

7. A 22-year-old man comes to the physician because of a 4-month history of recurrent episodes of muscle cramps and muscle weakness. He says the symptoms worsen soon after exercise which limits his endurance. He has no major illness and takes no medications. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show low serum potassium, metabolic alkalosis, and mildly elevated serum calcium. Genetic studies show a defective gene coding sodium-chloride symporter in the distal convoluted tubules. Which of the following is the most likely diagnosis in this patient?

A. Conn syndrome
B. Bartter syndrome
C. Gitelman syndrome
D. Liddle syndrome
E. Nephrogenic diabetes insipidus



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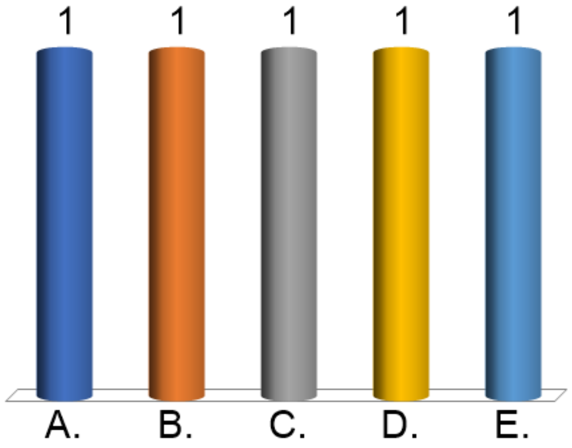
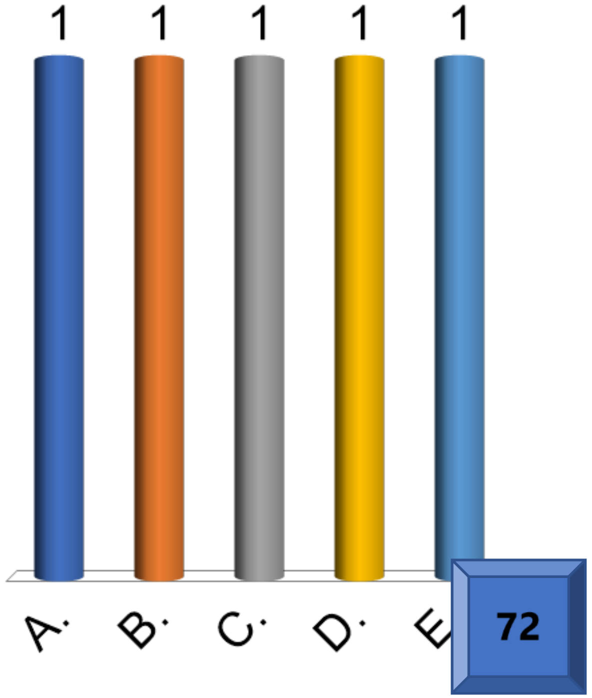


SOM.MK.II.BPM1.3.CPR.2.PHYS.0287. Describe the cellular mechanisms for the transport of Na+, Cl-, K+, HCO3-, Ca+2, phosphate, organic solutes (e.g., glucose, amino acids, and urea), and water by the major tubular segments.

8. A 64-year-old woman comes to the physician for a follow-up examination. She has a 15-year history of hypertension and type 2 diabetes mellitus and is adherent with anti-hypertensive and anti-diabetic medications. Her blood pressure is 136/88 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. Urinalysis shows trace amounts of proteins and glycosuria. Results of renal studies are shown: P_{GC} = 62 mmHg, P_{BS} = 30 mmHg, π_{GC} = 25 mmHg, π_{BS} = 3 mmHg, and K_f = 12 mL/min/mmHg.

Which of the following values best approximates the glomerular filtration rate (GFR) in this patient?

A. 70 mL/min
B. 85 mL/min
C. 95 mL/min
D. 100 mL/min
E. 120 mL/min



8. A 64-year-old woman comes to the physician for a follow-up examination. She has a 15-year history of hypertension and type 2 diabetes mellitus and is adherent with anti-hypertensive and anti-diabetic medications. Her blood pressure is 136/88 mm Hg; other vital signs are within normal limits. Physical examination shows no abnormalities. Urinalysis shows trace amounts of proteins and glycosuria. Results of renal studies are shown: P_{GC} = 62 mmHg, P_{BS} = 30 mmHg, π_{GC} = 25 mmHg, π_{BS} = 3 mmHg, and K_f = 12 mL/min/mmHg.

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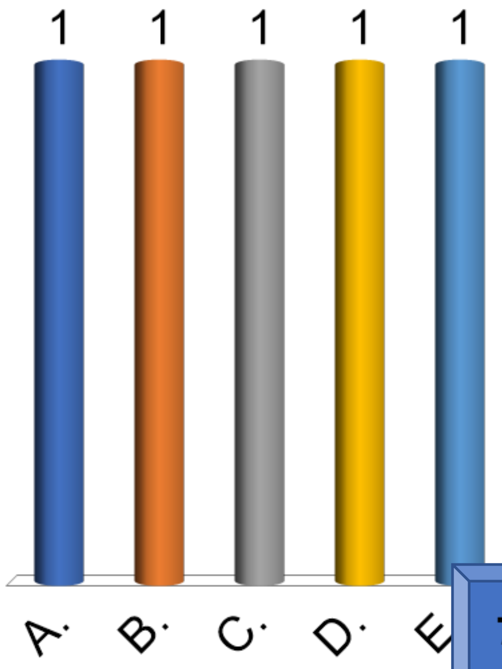
$$NFP = (P_{GC} + \pi_{BS}) - (P_{BS} + \pi_{GC})$$
$$= (62+3) - (30+25) = +10 \text{ mmHg}$$

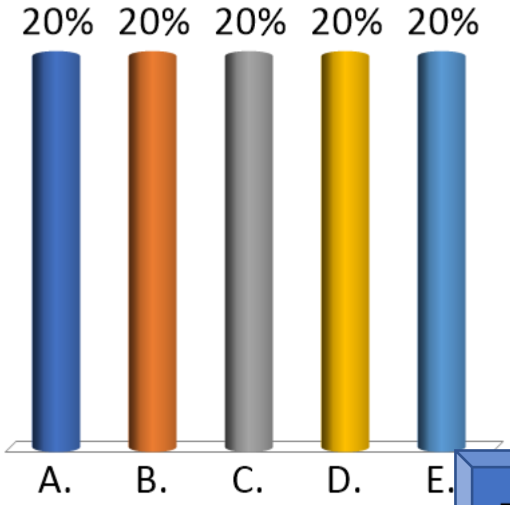
$$GFR = K_f \times NFP$$
$$= 12 \times 10 = 120$$



SOM.MK.II.BPM1.3.CPR.2.PHYS.0278. Given the capillary and Bowman's capsule hydrostatic and oncotic pressures, calculate the net filtration force at the glomerular capillaries.

9. A 41-year-old woman comes to the physician because of a 5-month history of recurrent episodes of headaches. She has a 6-year history of hyperlipidemia and hypertension and is non-adherent with medications. Her blood pressure is 165/98 mm Hg; other vital signs are within normal limits. Physical examination shows a loud S2 in the aortic area. Serum studies show elevated concentrations of renin, angiotensin II and sodium, but low concentration of potassium. Renal angiography shows bilateral renal artery stenosis. Which of the following additional laboratory alterations is most likely in this patient?

A. Decreased urinary excretion of potassium
B. Increased serum aldosterone
C. Increased urinary excretion of sodium
D. Increased serum anti-diuretic hormone
E. Decreased serum atrial natriuretic peptide



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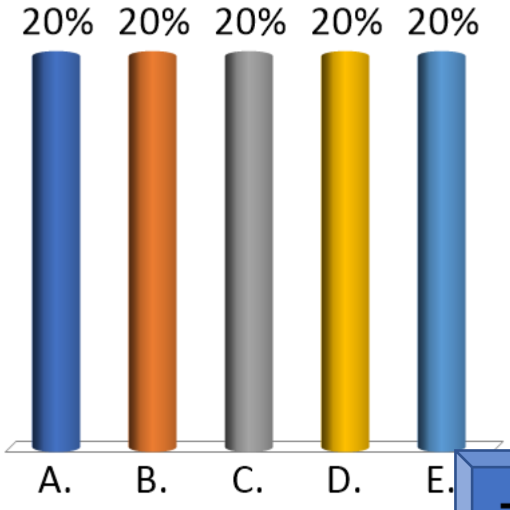
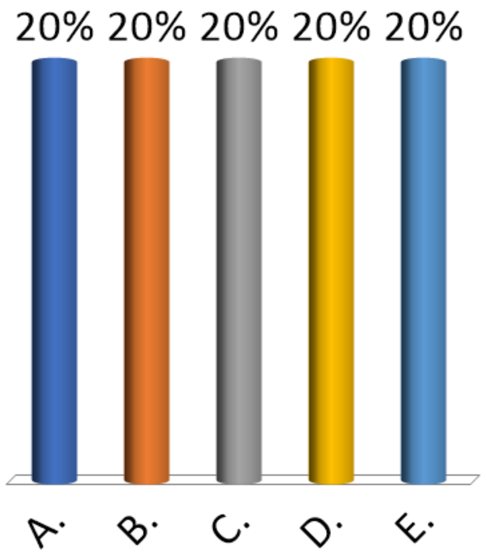


SOM.MK.II.BPM1.3.CPR.2.PHYS.0292. Outline the effects (up to five) of hormones on the various segments of the nephron.

10. A 42-year-old man comes to the physician because of a 6-month history of increased urination, thirst, and appetite. He is 5 ft. 6 in (168 cm) tall and weighs 89 kg (196 lb); BMR is 31.6 kg/m2 (N: 19-25). His vital signs are within normal limits. Physical examination shows central obesity and acanthosis nigricans in the axilla and neck. Results of laboratory studies are shown:
Blood glucose concentration = 350 mg/dL,
Glomerular filtration rate = 120 mL/min,
Transport maximum (T_m) of glucose = 360 mg/min. Assume that all nephrons have the same length. Which of the following values best approximates glucose excretion in urine in this patient?

A. 20 mg/min
B. 60 mg/min
C. 100 mg/min
D. 140 mg/min
E. 230 mg/min



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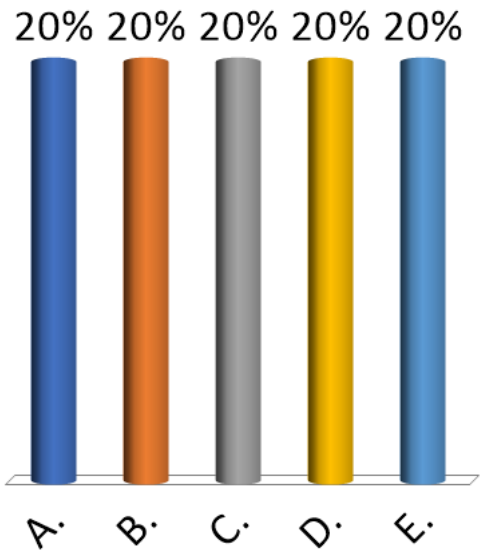
$FL=P_{glucose} \times GFR$
$=3.5 \times 120=420.$
$420-360= 60.$



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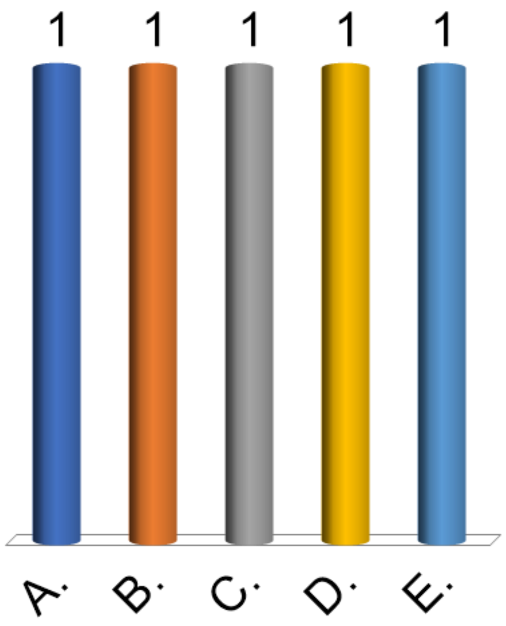
SOM.MK.II.BPM1.3.CPR.2.PHYS.0283. Define and understand the concepts of net transport: filtration, reabsorption, secretion, excretion.

11. Investigators conduct a study that evaluates the effect of aldosterone on sodium reabsorption in various parts of a nephron in 100 healthy volunteers. Using micropipettes, the investigators collect tubular fluids from different sites of the nephron and analyze their sodium concentrations. The serum concentration of aldosterone is maintained at a maximum concentration during the study. Which of the following sets of values best approximates the fraction of filtered load of sodium being reabsorbed under the maximum influence of the hormone studied?

Option	Proximal Convoluted Tubule	Loop of Henle	DCT and Collecting Duct
A	65%	10%	25%
B	10%	25%	65%
C	25%	10%	65%
D	65%	24%	10%
E	5%	5%	90%



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C	25%	10%	65%
✓D	65%	24%	10%
E	5%	5%	90%

SOM.MK.II.BPM1.3.CPR.2.PHYS.0290. For the major solutes (glucose, amino acids, Na, K, Ca, Mg, H2O) of the glomerular filtrate and for PAH and inulin, list the areas of the nephron where these molecules are reabsorbed and secreted.

12. A 30-year-old woman participates in a study related to the body fluids. Her vital signs are within normal limits. Physical examination shows no abnormalities. She is 150 cm (4 ft and 11 in) tall and weighs 52 kg (115 lb). Results of her body fluid studies show:

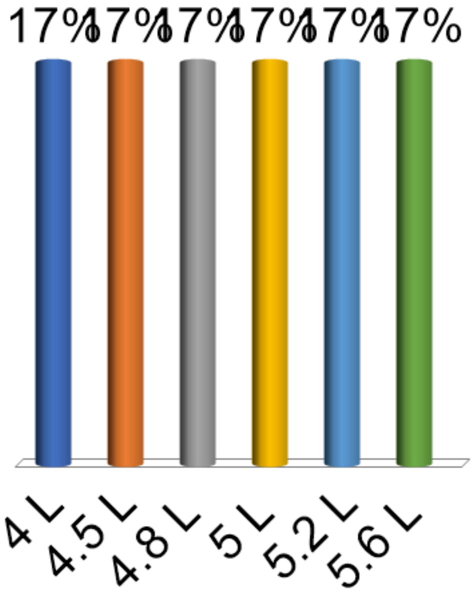
Inulin space = 16 L,

^{131}I-albumin space = 3.0 L,

Hematocrit = 0.375.

Which of the following values in liters (L) is the best estimation of her blood volume?

A. 4 L
B. 4.5 L
C. 4.8 L
D. 5 L
E. 5.2 L
F. 5.6 L



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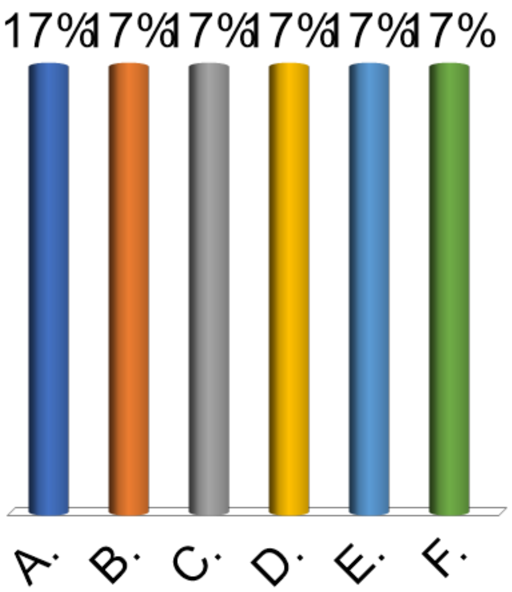
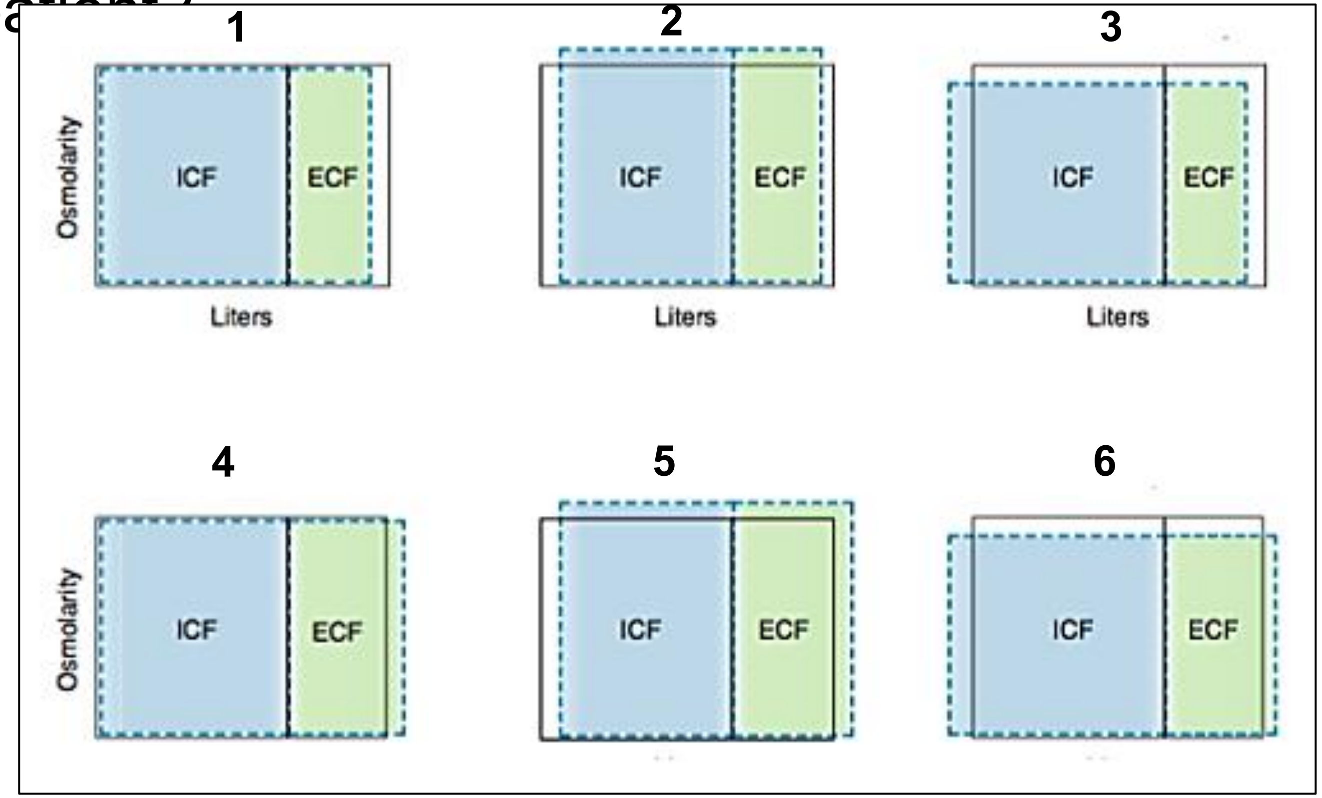
Blood volume = Plasma volume/(1-Hematocrit)
= 3/(1-0.375) = 3/0.625
= 3000/625 = 4.8

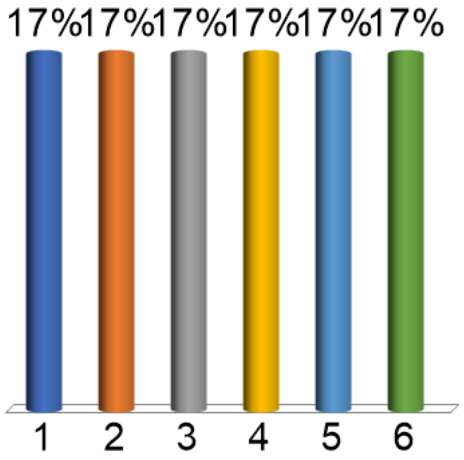


SOM.MK.II.BPM1.3.CPR.2.PHYS.0295. Demonstrate the ability to use the indicator dilution principle to measure plasma volume, blood volume, extracellular fluid volume, and total body water, and identify compounds used to measure each volume.

13. A 10-year-old boy is brought to the emergency department by his mother because of a 1-day history of headaches and 2-hour history of confusion. The mother says he has eaten a large amount of chips and drank several liters of water in the last 3 days. His pulse is 96/min and blood pressure is 124/86 mm Hg. Physical examination shows altered mental status but no signs of dehydration. Laboratory studies show a serum sodium concentration of 170 mEq/L (Normal: 135-145). Which of the following Darrow-Yannet diagrams best represents this pa



A. 1
B. 2
C. 3
D. 4
E. 5
F. 6



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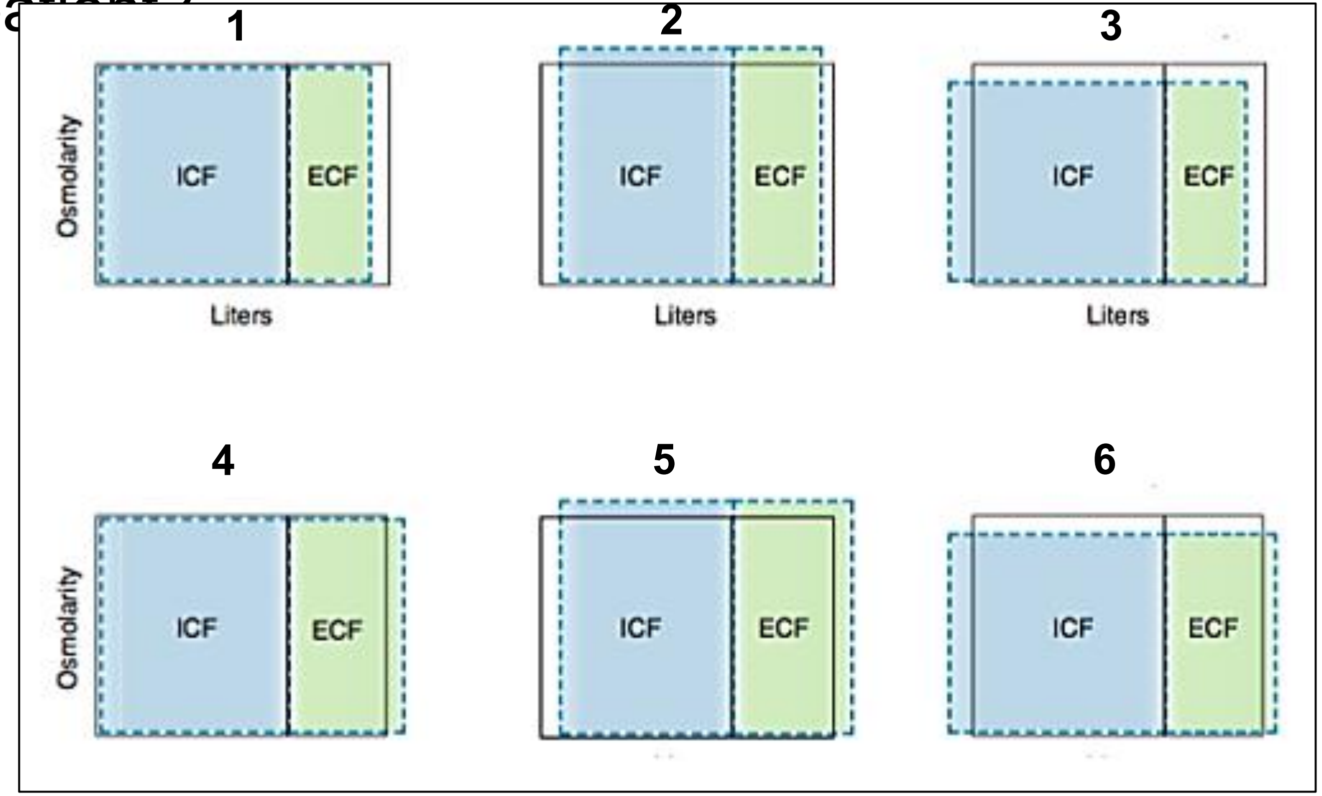
13. A 10-year-old boy is brought to the emergency department by his mother because of a 1-day history of headaches and 2-hour history of confusion. The mother says he has eaten a large amount of chips and drank several liters of water in the last 3 days. His pulse is 96/min and blood pressure is 124/86 mm Hg. Physical examination shows altered mental status but no signs of dehydration. Laboratory studies show a serum sodium concentration of 170 mEq/L (Normal: 135-145). Which of the following Darrow-Yannet diagrams best represents this patient?

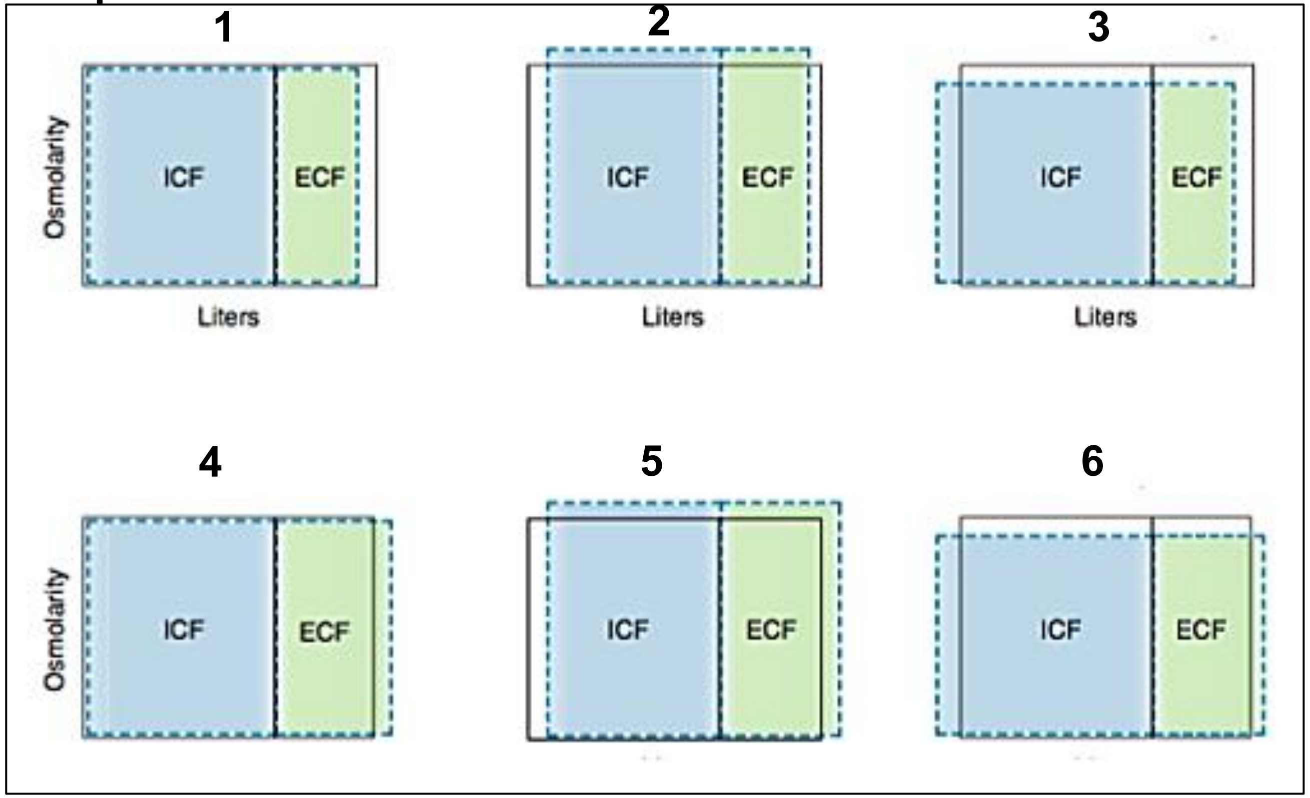
A. 1
B. 2
C. 3
D. 4
✓E. 5
F. 6

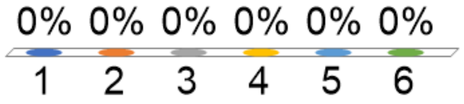


SOM.MK.II.BPM1.3.CPR.2.PHYS.0296.Explain the water movement between intracellular and extracellular fluid compartments caused by increase or decrease in extracellular fluid osmolality.

14. A 41-year-old woman comes to the physician because of a 5-week history of muscle weakness, fatigue, and increased skin pigmentation. Her blood pressure is 110/64 mm Hg. Physical examination shows hyperpigmentation of skin and sunken eyes. Laboratory studies show low serum sodium, glucose, and osmolarity but increased serum potassium and bicarbonate. Body fluid estimation shows a low extracellular fluid volume. Which of the following Darrow-Yannet diagrams best represents this patient?

A. 1
B. 2
C. 3
D. 4
E. 5
F. 6





72

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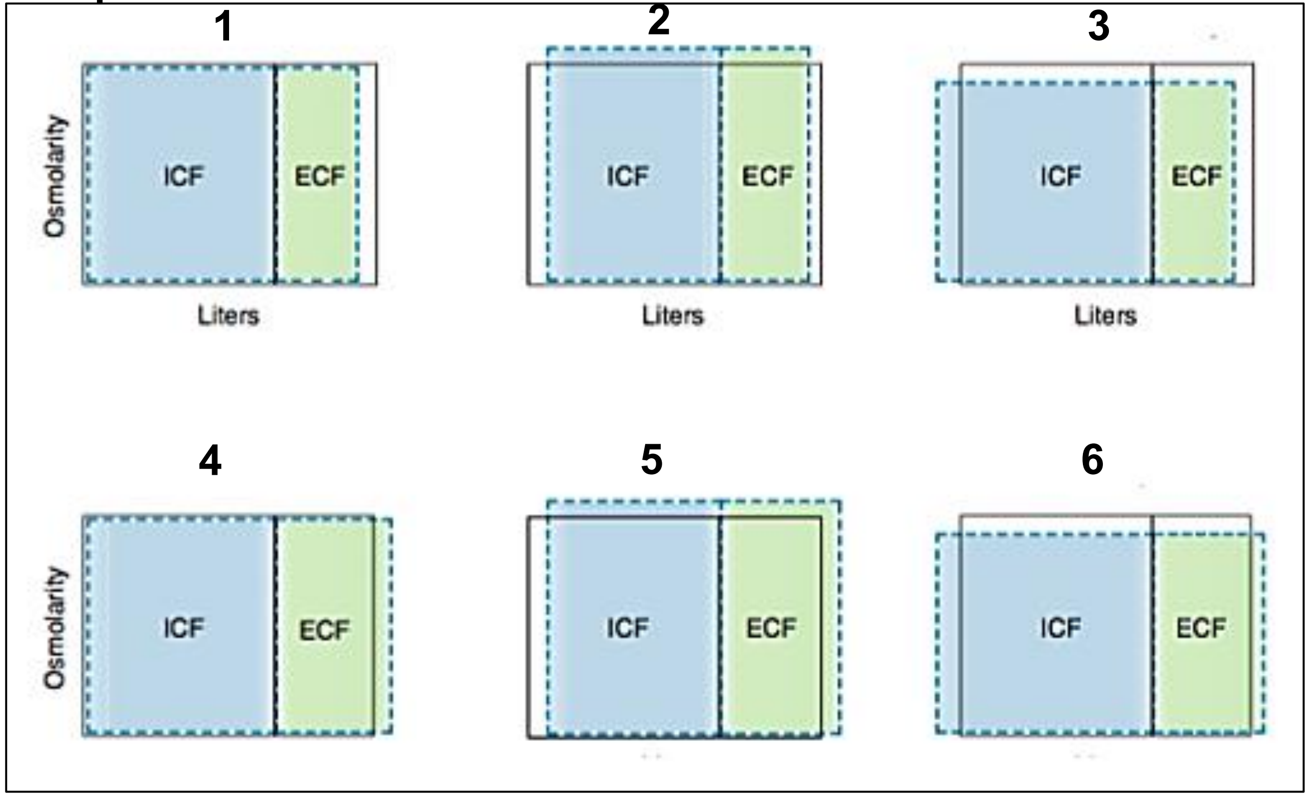
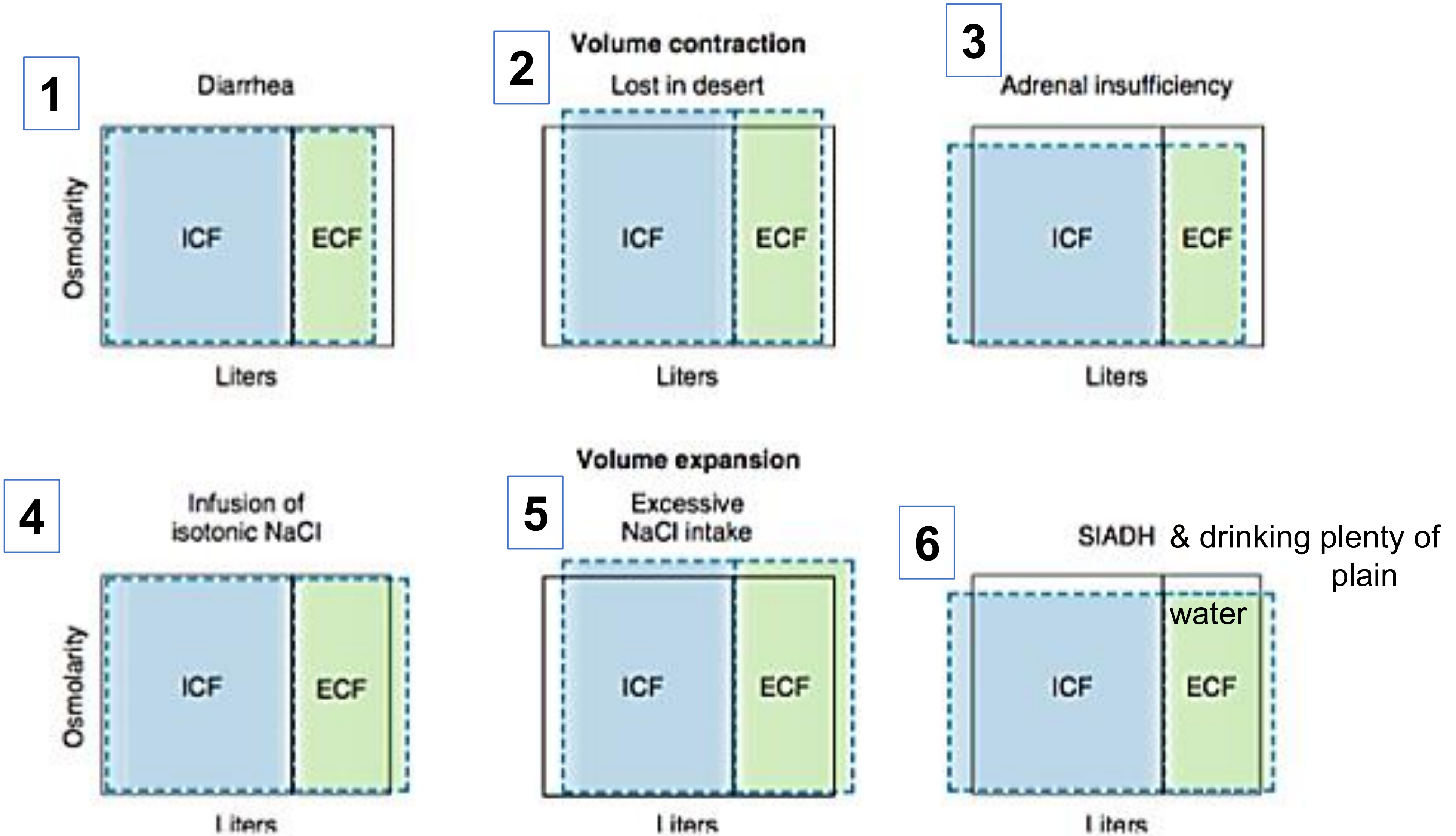


SOM.MK.II.BPM1.3.CPR.2.PHYS.0296.Explain the water movement between intracellular and extracellular fluid compartments caused by increases or decreases in extracellular fluid osmolality.

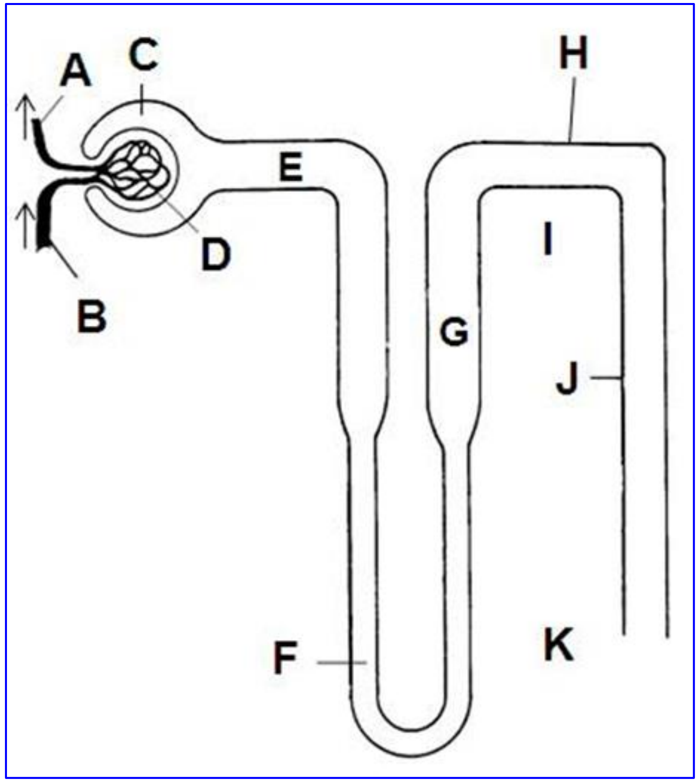
Darrow-Yannet diagrams in different clinical conditions

Figures 1, 2 & 3: Volume contractions

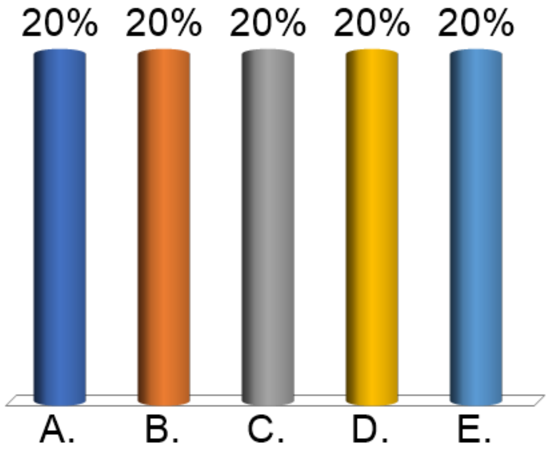
Figures 4, 5 & 6: Volume expansions



15. A 36-year-old policeman is brought to the emergency department 15 minutes after he collapsed while working at a busy traffic site. His colleague says he has been working for the past 10 hours and had minimal fluid intake. His pulse is 116/min and blood pressure is 106/64 mm Hg. Physical examination shows altered mental status, dry mucous membranes, and decreased skin turgor. Laboratory studies show elevated plasma vasopressin (ADH) concentration. A diagram of the nephron is shown with labels A through K. Which of the following sets of osmolarities is most likely in this patient?



Option	Osmolarity at site E	Osmolarity at site K
A	Hypo-osmotic	Hyperosmotic
B	Hypo-osmotic	Isosmotic
C	Isosmotic	Hyperosmotic
D	Hyperosmotic	Hypo-osmotic
E	Isosmotic	Hypo-osmotic



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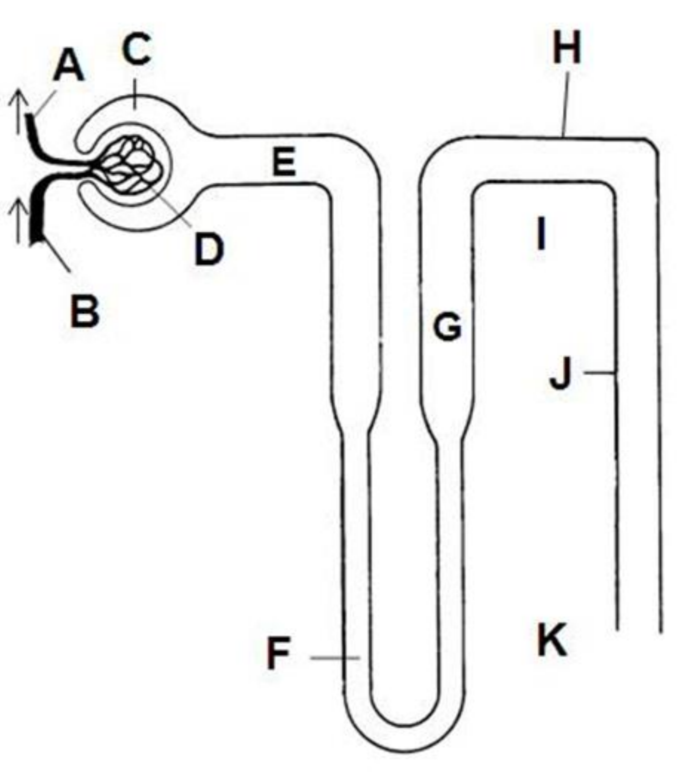

Option	Osmolarity at site E	Osmolarity at site K
A	Hypo-osmotic	Hyperosmotic
B	Hypo-osmotic	Isosmotic
✓ C	Isosmotic	Hyperosmotic
D	Hyperosmotic	Hypo-osmotic
E	Isosmotic	Hypo-osmotic

SOM.MK.II.BPM1.3.CPR.2.PHYS.0378. Beginning with the loop of Henle, compare and contrast the tubular fluid and interstitial fluid osmolarity changes that allow either a dilute or concentrated urine to be produced and excreted.

16. A 37-year-old woman is brought to the physician because of a 3-day history of vomiting, diarrhea, and abdominal pain. She recently returned from a trip to Brazil where she ate some street food. Her blood pressure is 110/70 mm Hg; other vital signs are within normal limits. Physical examination shows dry mucous membranes and poor skin turgor. Laboratory studies show increased serum osmolarity and vasopressin concentration but normal serum glucose concentration. A 24-hour urine volume is reduced. Which of the following sets of renal tubular functions is most likely expected in this patient?

Option	Water reabsorption in collecting ducts	Urea reabsorption in outer medullary collecting ducts	Urea reabsorption in inner medullary collecting ducts	
A	Increases	No change	Increases	1
B	Increases	Increases	No change	1
C	No change	Increases	Increases	1
D	Increases	Increases	Increases	1
E	No change	No change	Increases	1

16. A 37-year-old woman is brought to the physician because of a 3-day history of vomiting, diarrhea, and abdominal pain. She recently returned from a trip to Brazil where she ate some street food. Her blood pressure is 110/70 mm Hg; other vital signs are within normal limits. Physical examination shows dry mucous membranes and poor skin turgor. Laboratory studies show increased serum osmolarity and vasopressin concentration but normal serum glucose concentration. A 24-hour urine volume is reduced. Which of the following sets of renal tubular functions is most likely expected in this patient?

Option	Water reabsorption in collecting ducts	Urea reabsorption in outer medullary collecting ducts	Urea reabsorption in inner medullary collecting ducts	
A	Increases	No change	Increases	1
B	Increases	Increases	No change	1
C	No change	Increases	Increases	1
D	Increases	Increases	Increases	1
E	No change	No change	Increases	1

SOM.MK.II.BPM1.3.CPR.2.PHYS.0327. Explain the tubular section and cellular mechanism by which ADH increases permeability to water and urea.